NIKHIL SAINI

Mumbai, India | +91 9145990892 | nikhil1010saini@gmail.com
Portfolio | LinkedIn | GitHub | LeetCode 2300+ | Codeforces 1980+ | CodeChef 2000+

EDUCATION

Indian Institute of Technology Bombay **CPI: 7.89/10**
Dual Degree (B.Tech.+M.Tech.), ***Mechanical Engineering*** *- CAD & Automation; Minor:* ***Electrical Engg.*** *2020 - 2025*

- Awarded the academic distinction of **AA grade** in total of **11 courses** over the span of 10 semesters
- Scored a perfect **100 percentile** in Mathematics in **JEE Mains** and ranked among the top **3 percentile** in **JEE Advanced**

TECHNICAL PROJECTS

Discord-Style Chat Platform [Live] | *Next.js, TypeScript, Tailwind, Prisma, SQL, WebSockets, LiveKit, Clerk* **Aug'25-Sep'25**

- Engineered a **full-stack real-time** chat platform with text, voice, and video channels using **WebSockets** and **LiveKit**
- Implemented **role-based access control** and secure authentication with **Clerk**, backed by a **Prisma relational data layer**
- Built a **responsive UI** with dark/light themes, **infinite message loading**, and media sharing via **ShadCN** and **UploadThing**

Kaali Tirri Web Multiplayer Game [Live] | *WebRTC, PeerJS, STUN/TURN servers, Web Audio API* **Sep'26-Oct'26**

- Architected a **serverless WebRTC** multiplayer engine using STUN/TURN protocols for **real-time state** synchronization
- Secured the P2P data layer with **strict state sanitization** against memory exploits and **rate-limiting** to mitigate network spam
- Engineered seamless **session recovery** and host migration via **JSON state persistence**, peer ID remapping, and **heartbeat pings**

AI-First CRM – Conversational Interaction Logger | *FastAPI, LangGraph, LangChain, React, Redux, Groq* **Jun'26**

- Built an AI-first CRM that lets field reps **log customer interactions** through a **conversational agent** avoiding manual forms
- Designed an autonomous **LangGraph** tool-calling agent with **6 tools** orchestrating a Groq-hosted **Llama 3.3 70B model**
- Synchronized dynamic agent-extracted form fields to a **React/Redux** UI in **real time** utilizing **FastAPI** and **SQLite** storage

Image-Based Sudoku Solver | *Python, TensorFlow, OpenCV* **May'24**

- Programmed an image-based Sudoku solver using **TensorFlow** and **OpenCV**, extracting puzzles from the images
- Optimized Sudoku **digit preprocessing** with blurring, thresholding, and noise removal, achieving **98%** MNIST accuracy
- Implemented a robust **backtracking algorithm** to solve a Sudoku in **0.6 ms**, showcasing consistent and reliable result

Kanban Board Web App | *React.js, REST API, Axios* **Mar'25**

- Built an **interactive** board in React to **visualize and manage tasks** dynamically with grouping by **status, user and priority**
- **Persisted** user-selected grouping and sorting configurations via **LocalStorage**, improving user experience across sessions
- Designed a **responsive UI** with custom CSS & Integrated **RESTful API** data using **Axios**, enabling **real-time** rendering of tickets

Budget Tracker App | *Dart, Flutter, Firestore, Firebase* **May'23-Jul'23**

- Developed an interactive and responsive user interface using **Flutter** to **track** & **analyze** their daily/monthly expenditure
- Integrated **Firestore** for data management and **Firebase** for **authentication** to ensure secure & efficient user interactions

EXPERIENCE & LEADERSHIP

Teaching Assistant | *Statistical Machine Learning & Data Mining* | Prof. Asim Tewari **Aug'24-Nov'24**

- Assisted a professor in overseeing a class of **300+** students by resolving doubts, managing assignments & examinations
- Coordinated with a team of **17** Teaching Assistants to manage daily class quizzes, ensuring fair and timely assessments

Project Manager | Enactus, IIT Bombay **May'23-May'24**

- Created a **responsive website** showcasing 4+ social entrepreneurship projects for Enactus IITB utilizing **ReactJS**
- Edited project video of **3+ minutes** for the **National Enactus Competition**, synthesizing prior team milestones

TECHNICAL SKILLS

Languages: **C++, Python, JavaScript/TypeScript**, Java, **SQL**, HTML/CSS, Dart
Frameworks & Libraries: **React JS, Next JS, Node JS, Redux**, React Native, **Tailwind, ShadCN**, Flutter, **Prisma**
Databases & Cloud: **MongoDB**, Firebase/Firestore, **SQLite**, GFG AWS (Cloud Practitioner & Solutions Architect)
Testing & Tools: Selenium, TestNG, Cucumber BDD, Maven, Jenkins, REST Assured, **Git, Postman, Figma**
Certifications (GFG): React JS Development, Next JS Development, Full Stack Dev Bootcamp, CS Core (OS/DBMS/CN)
Competitive Programming: LeetCode, **Guardian (2300+ Rating, 1055+ solved)** · Codeforces, **Candidate Master (1980+** Rating, 243+ solved) · CodeChef, **5-Star (2000+ Rating, 880+ solved)**